

# 2D RANS Simulation of an Axisymmetric Turbulent Round Jet

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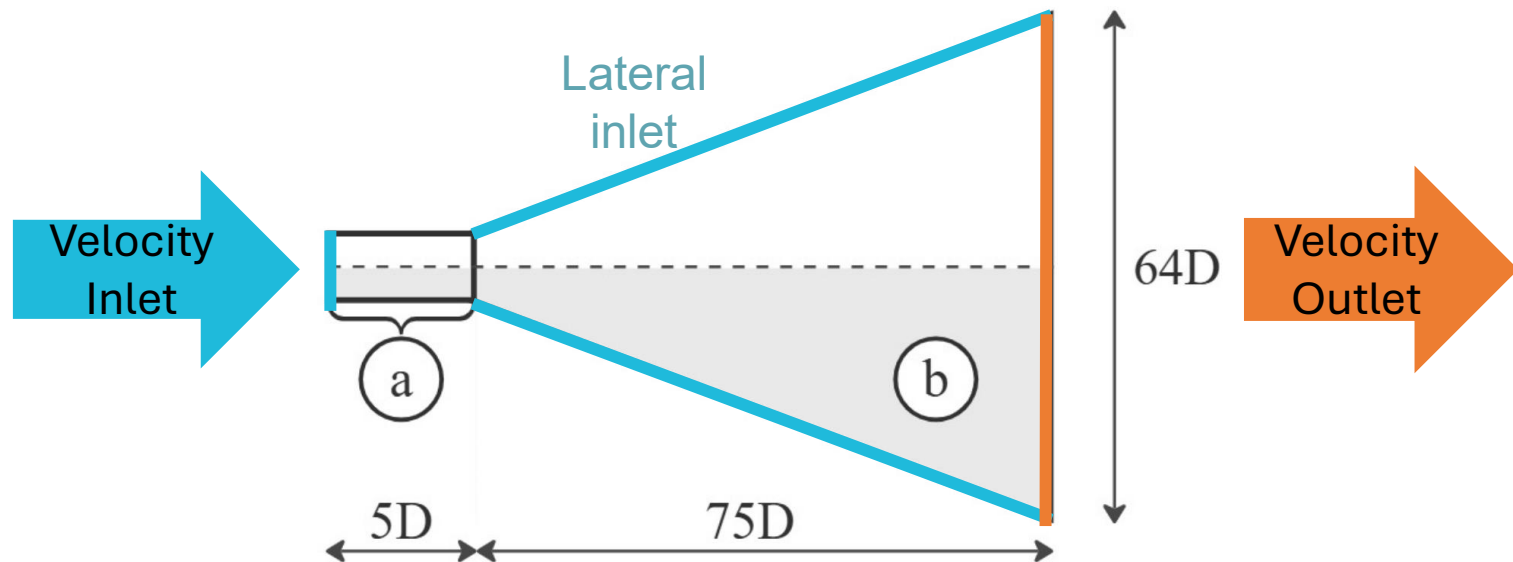


# Problem description & geometry

3D DNS Reference setup

- $Re = 3500$
- $D = 0.2$  m

a) Turbulent tube inlet  
b) Study domain

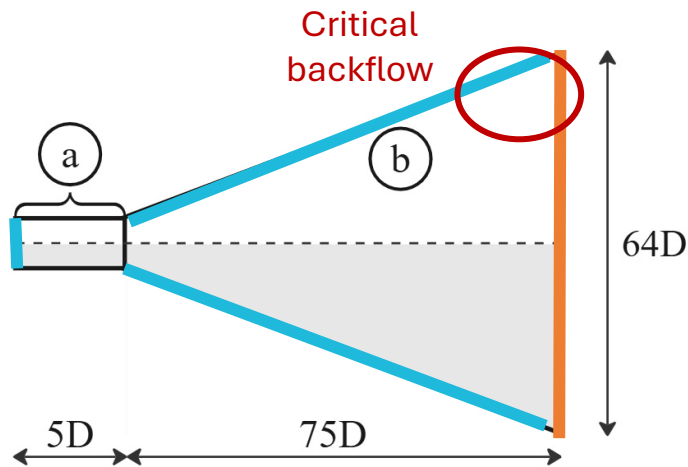




# CFD modelling strategy

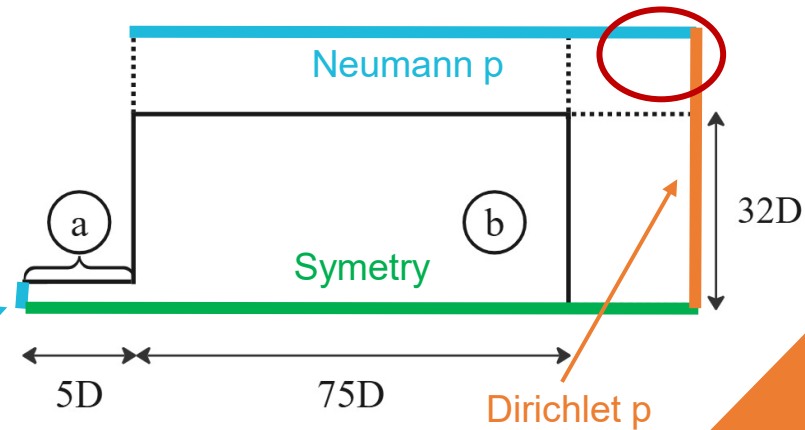
## 2D RANS CFD Model

- Domain and boundary readapted
- Multibloc structured mesh with quadrilateral elements



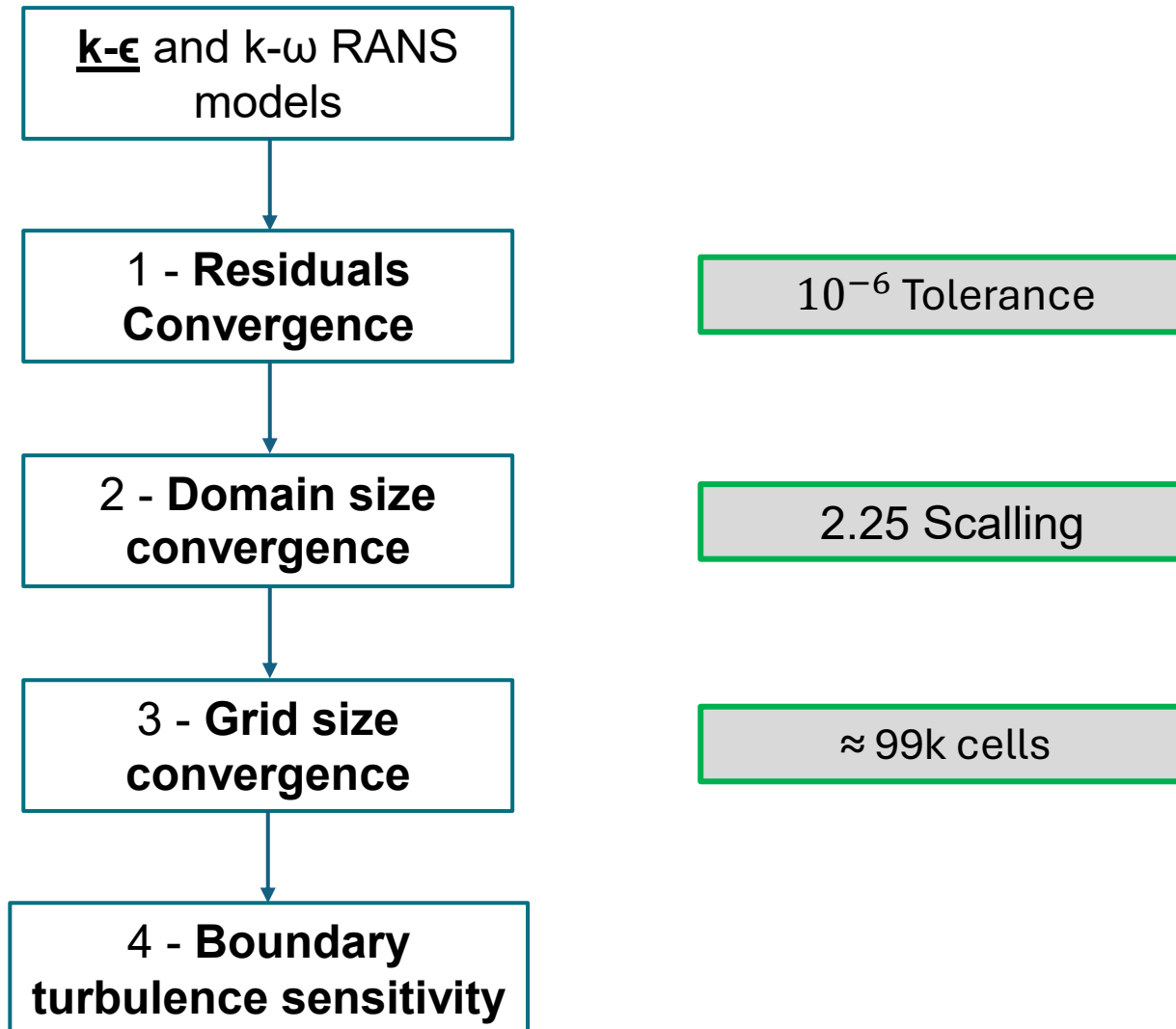
Ansys Fluent implementation

Dirichlet elliptic U





# Convergence / Verification process





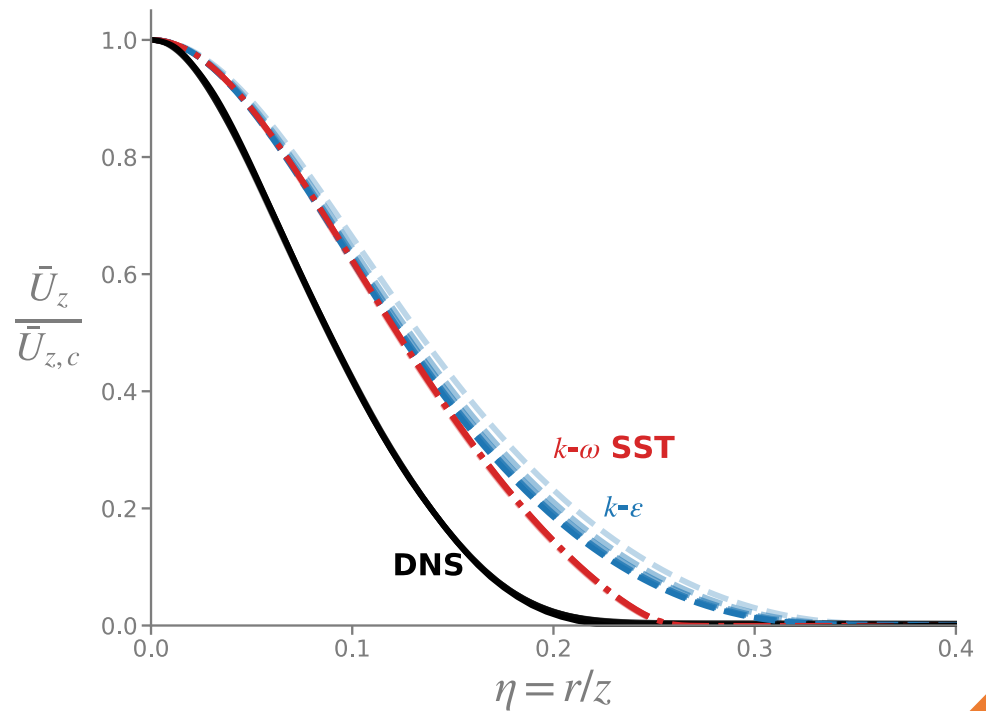
# Results : Self-Similarity

$z/D \geq 25$  : all normalized quantities  
collapse on one curve



Only functions of  $\eta$

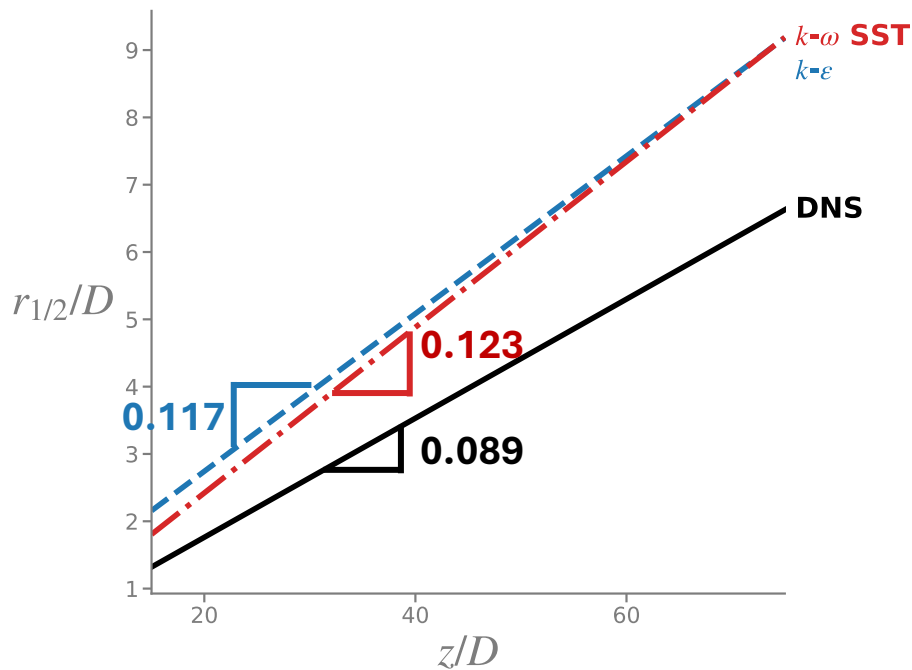
Self-similarity captured by  
RANS models



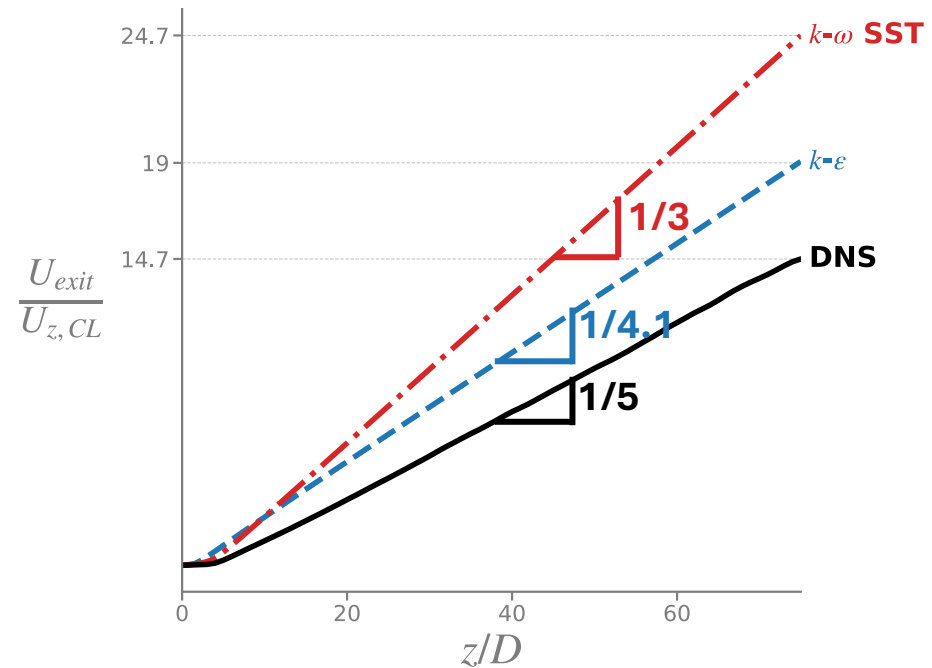


# Results : The Round Jet Anomaly

How it fails : Spreading Rate and Velocity Decay



Spreading rate overestimated  
by 30 - 40 %

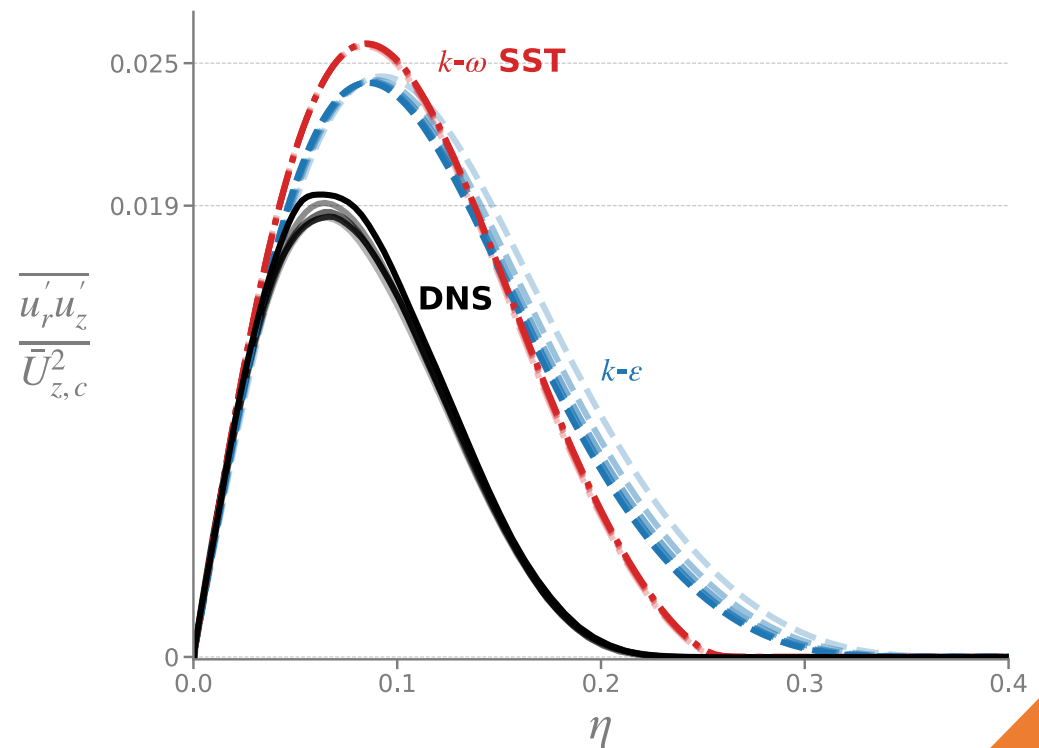
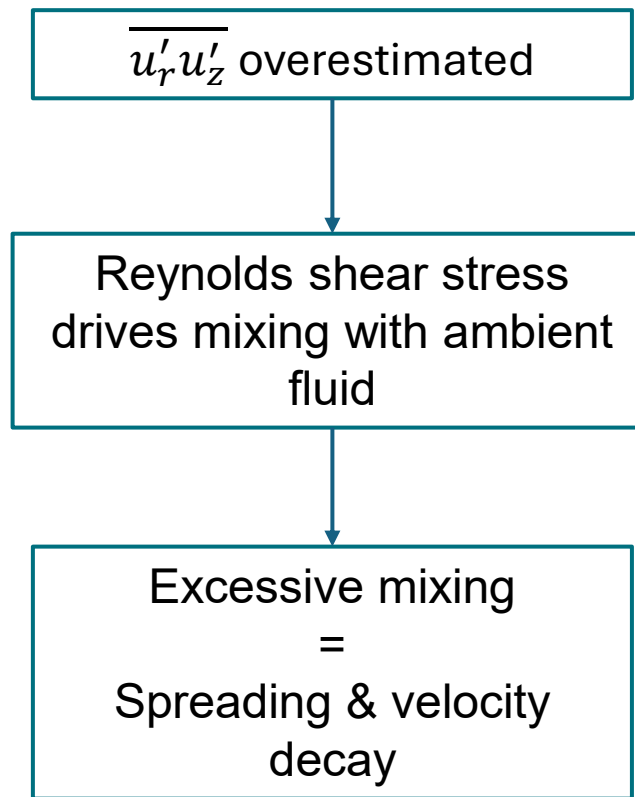


Decay constant  
underestimated by 20 - 40 %



# The Mechanism of the Anomaly

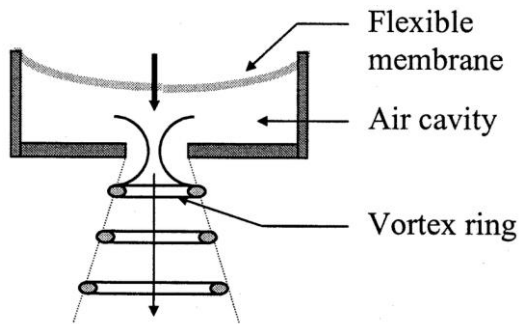
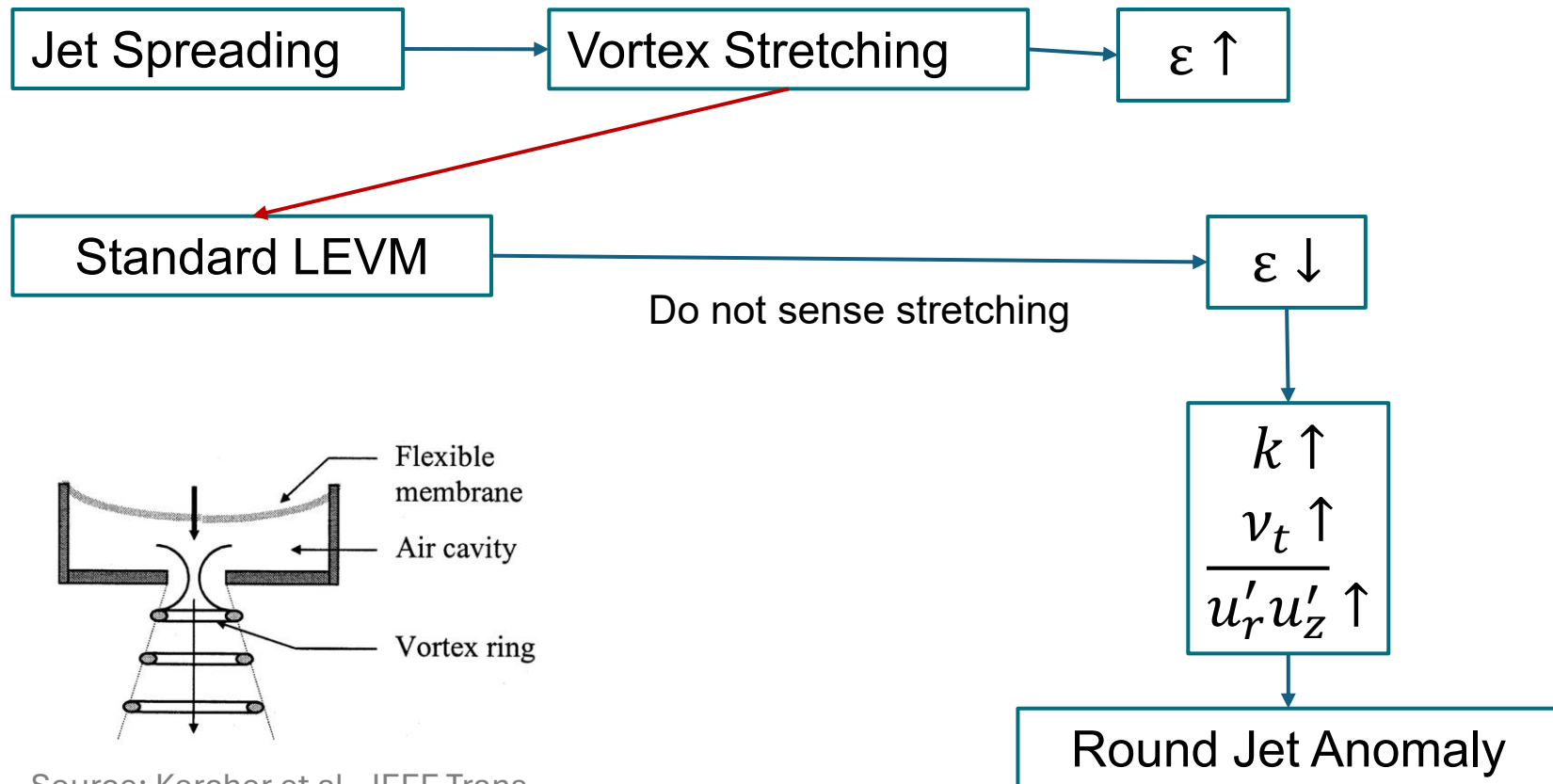
## Reynolds Shear Stress and Eddy Viscosity





# Why it Fails: Vortex Stretching

## Dissipation Deficit

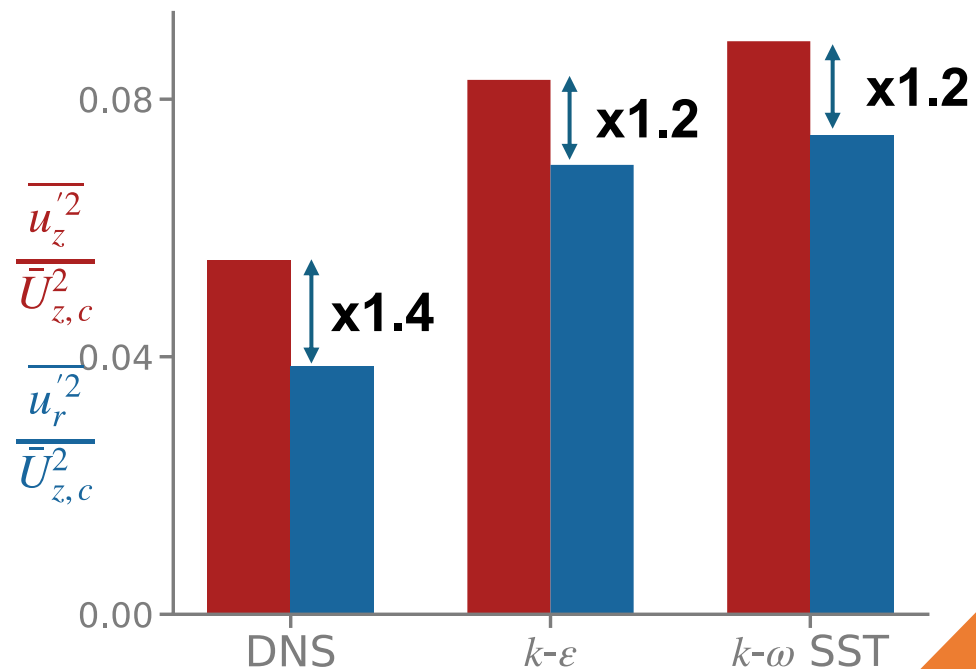
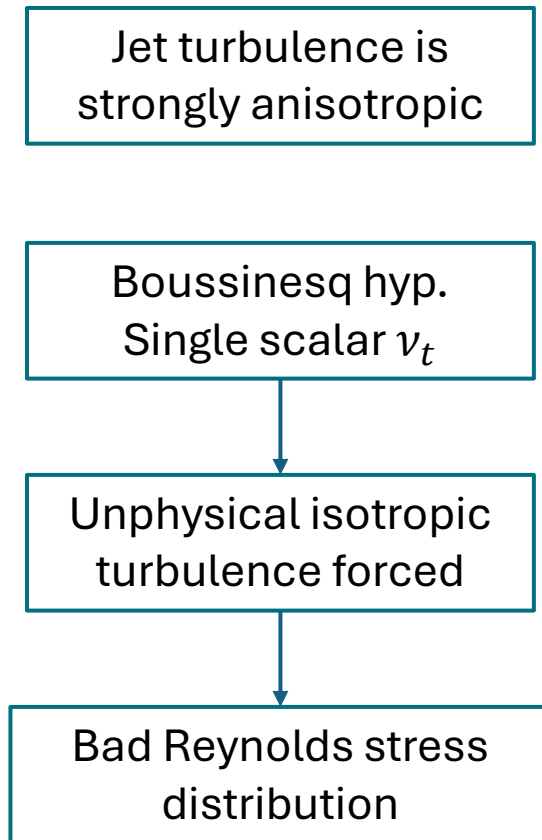


Source: Kercher et al., IEEE Trans. Comp. Packag. Technol., 2003



# Why it Fails: Turbulence Anisotropy

## Limitations of the Boussinesq Hypothesis





# Conclusion

- **Summary:** RANS models are qualitatively good but quantitatively inaccurate. Large overprediction of jet spreading for both models
- **Future Work:** Use Pope\* correction term in  $\varepsilon$ -equation for vortex stretching, Reynolds Stress Models (RSM) for anisotropy or LES/DES for higher-fidelity

\* *An explanation of the turbulent round jet/plane jet anomaly*, S. Pope, 1978



Thank you for your  
attention !

